Human Obstacle Detector and Tracker for Acme Robotics

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<tr><th>1. OVERVIEW & OBJECTIVES</th><th>2. APPROACH & METHODOLOGY</th></tr>
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Project: Human Obstacle Detector and Tracker Duration: 3 weeks Core Objectives:

Detect humans in real-time (YOLOv8)
Track persistent IDs for each human
Estimate 3D position in robot-centric frame
Deliver a TDD module with 90%+ code coverage

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Development Process:

Agile Iterative Process (AIP)
Test-Driven Development (TDD)
Pair Programming with role rotation
CI/CD with GitHub Actions & CodeCov

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<tr><th>3. TECHNICAL DETAILS</th><th>4. SCHEDULE & DELIVERABLES</th></tr>
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Technologies:

C++17, CMake, OpenCV 4.x
Google Test (GTest), cppcheck, Doxygen
ONNX Runtime (for DNN inference)
GitHub Actions (CI), CodeCov (Coverage)

Key Algorithms:

DNN-based 2D Detection (YOLOv8)
Intersection-over-Union (IoU) Tracking
Monocular Depth Estimation (MiDaS)
2D-to-3D Pinhole Camera Projection

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Project Schedule:

Wk 1 (Phase 0): Design, UML, Documentation
Wk 2 (Phase 1): Core Implementation (Sprints 1-2)
Wk 3 (Phase 2): Integration & Testing (Sprint 3)

Key Deliverables:
✓ UML diagrams & design documentation
✓ CI/CD pipeline with 90%+ test coverage
✓ Production-ready C++ module
✓ Doxygen-generated developer API
✓ Final demonstration video
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